

Homework 8 - MA 513 - Due Thu 2026-4-18

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1

$$f(z) = \frac{1}{z(z+1)} = \frac{A}{(1-z)} + \frac{B}{z}$$

$$\begin{aligned} 1 &= (-A + B)z + B \\ B &= 1 \quad A = 1 \end{aligned}$$

$$f(z) = \frac{1}{(1-z)} + \frac{1}{z}$$

$$f(z) = z^{-1} + \sum_{n=0}^{\infty} z^n$$

$$f(z) = \sum_{n=-1}^{\infty} z^n.$$

2

2.1 a)

$$\begin{aligned}
f(z) &= \frac{1}{z(z^2 + 1)(-z^2 + 4)} \\
&= \frac{-1}{z(z-i)(z+i)(z^2-4)} = \frac{A}{z} + \frac{B}{z-i} + \frac{C}{z+i} + \frac{D}{z-2} + \frac{E}{z+2} \\
-1 &= A((z-i)(z+i)(z^2-4)) + B(z(z+i)(z^2-4)) + C(z(z-i)(z^2-4)) \\
&\quad + D(z(z-i)(z+i)(z+2)) + E(z(z-i)(z+i)(z-2)) \\
-1 &= A(z^4 - 3z^2 - 4) + B(z^4 + iz^3 - 4z^2 - 4iz) + C(z^4 - iz^3 - 4z^2 + 4iz) \\
&\quad + D(z^4 + 2z^3 + z^2 + 2z) + E(z^4 - 2z^3 + z^2 - 2z) \\
-1 &= z^4(A + B + C + D + E) + z^3(iB - iC + 2D - 2E) \\
&\quad + z^2(-3A - 4B - 4C + D + E) + z(-4iB + 4iC + 2D - 2E) - 4A \\
A = \frac{1}{4} \quad B = C = -\frac{1}{10} \quad D = E = -\frac{1}{40} \\
&= \frac{1}{4z} - \frac{1}{10(z-i)} - \frac{1}{10(z+i)} - \frac{1}{40(z-2)} - \frac{1}{40(z+2)} \\
&= \frac{1}{4}z^{-1} + \frac{1}{10i(1-\frac{z}{i})} - \frac{1}{10i(1-\frac{-z}{i})} + \frac{1}{80(1-\frac{z}{2})} - \frac{1}{80(1-\frac{-z}{2})} \\
&= \frac{1}{4}z^{-1} - \frac{i}{10} \sum_{n=0}^{\infty} \left(\frac{z}{i}\right)^n + \frac{i}{10} \sum_{n=0}^{\infty} \left(\frac{-z}{i}\right)^n + \frac{1}{80} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n - \frac{1}{80} \sum_{n=0}^{\infty} \left(\frac{-z}{2}\right)^n \\
&= \frac{1}{4}z^{-1} + \sum_{n=0}^{\infty} \frac{i}{10} ((iz)^n - (-iz)^n) + \frac{1}{80} \left(\left(\frac{z}{2}\right)^n - \left(\frac{-z}{2}\right)^n \right) \\
&= \frac{1}{4}z^{-1} + \sum_{n=0}^{\infty} \left(\frac{i}{10} (i)^n - (-i)^n + \frac{1}{80} \left(\left(\frac{1}{2}\right)^n - \left(\frac{-1}{2}\right)^n \right) \right) z^n
\end{aligned}$$

2.2 b)

$$\begin{aligned}
f(z) &= \frac{1}{z(z^2 + 1)(-z^2 + 4)} \\
&= \frac{1}{4z} - \frac{1}{10(z - i)} - \frac{1}{10(z + i)} - \frac{1}{40(z - 2)} - \frac{1}{40(z + 2)} \\
&= \frac{1}{4} \frac{1}{1 - (-z + 1)} - \frac{1}{10(1 - i)} \frac{1}{(1 - \frac{-z+1}{1-i})} - \frac{1}{10(1 + i)} \frac{1}{1 - (\frac{-z+1}{1+i})} + \frac{1}{40} \frac{1}{1 - (z - 1)} - \frac{1}{120} \frac{1}{1 - (\frac{-z+1}{3})} \\
&= \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n (z - 1)^n - \frac{1}{10(1 - i)} \sum_{n=0}^{\infty} \left(\frac{-1}{1 - i}\right)^n (z - 1)^n - \frac{1}{10(1 + i)} \sum_{n=0}^{\infty} \left(\frac{-1}{1 + i}\right)^n (z - 1)^n + \frac{1}{40} \sum_{n=0}^{\infty} (z - 1)^n \\
&= \sum_{n=0}^{\infty} \left(\frac{1}{4} (-1)^n - \frac{1}{10(1 - i)} \left(\frac{-1}{1 - i}\right)^n - \frac{1}{10(1 + i)} \left(\frac{-1}{1 + i}\right)^n + \frac{1}{40} - \frac{1}{120} \left(\frac{-1}{3}\right)^n \right) (z - 1)^n
\end{aligned}$$

2.3 c)

$$\begin{aligned}
f(z) &= \frac{1}{z(z^2 + 1)(-z^2 + 4)} \\
&= \frac{1}{4z} - \frac{1}{10(z - i)} - \frac{1}{10(z + i)} - \frac{1}{40(z - 2)} - \frac{1}{40(z + 2)} \\
&= \frac{1}{8} \frac{1}{1 - (\frac{-z+2}{2})} - \frac{1}{10(2 - i)} \frac{1}{1 - (\frac{-z+2}{2-i})} - \frac{1}{10(2 + i)} \frac{1}{1 - (\frac{-z+2}{2+i})} \\
&\quad - \frac{1}{40}(z - 2)^{-1} - \frac{1}{160} \frac{1}{1 - (\frac{-z+2}{4})} \\
&= \frac{1}{8} \sum_{n=0}^{\infty} \left(\frac{-1}{2}\right)^n (z - 2)^n - \frac{1}{10(2 - i)} \sum_{n=0}^{\infty} \frac{1}{2 - i} \left(\frac{-1}{2 - i}\right)^n (z - 2)^n - \frac{1}{10(2 + i)} \sum_{n=0}^{\infty} \left(\frac{-1}{2 + i}\right)^n (z - 2)^n \\
&\quad - \frac{1}{40}(z - 2)^{-1} - \frac{1}{160} \sum_{n=0}^{\infty} \left(\frac{-1}{4}\right)^n (z - 2)^n \\
&= -\frac{1}{40}(z - 2)^{-1} + \sum_{n=0}^{\infty} \left(\frac{1}{8} \left(\frac{-1}{2}\right)^n - \frac{1}{10(2 - i)} \frac{1}{2 - i} \left(\frac{-1}{2 - i}\right)^n - \frac{1}{10(2 + i)} \left(\frac{-1}{2 + i}\right)^n \right. \\
&\quad \left. - \frac{1}{160} \left(\frac{-1}{4}\right)^n \right) (z - 2)^n
\end{aligned}$$

3

$$f(z) = \sum_{n=0}^{\infty} \frac{(1/z)^n}{n!} + \sum_{n=0}^{\infty} \frac{(z)^n}{n!}$$

$$f(z) = \sum_{n=0}^{\infty} \frac{(z)^{-n}}{n!} + \sum_{n=0}^{\infty} \frac{(z)^n}{n!}$$

$$f(z) = \sum_{n=-\infty}^{-1} \frac{(z)^n}{(-n)!} + 2 \frac{(z)^0}{0!} + \sum_{n=1}^{\infty} \frac{(z)^n}{n!}$$

$$f(z) = \sum_{n=-\infty}^{-1} \frac{(z)^n}{(-n)!} + 2 + \sum_{n=1}^{\infty} \frac{(z)^n}{n!}.$$

4

$$\text{Res}(f, -i) = 2\pi i \lim_{z \rightarrow -i} (z+i)f(z)$$

$$= 2\pi i \lim_{z \rightarrow -i} (z+i)$$

$$\cdot \frac{1}{(z-i)(z+i) \left(z - \left(\frac{\sqrt{3}}{3} + \frac{1}{2}\right)\right) \left(z + \left(\frac{\sqrt{3}}{3} + \frac{1}{2}\right)\right) \left(z - \left(\frac{\sqrt{3}}{3} - \frac{1}{2}\right)\right) \left(z + \left(\frac{\sqrt{3}}{3} - \frac{1}{2}\right)\right)}$$

$$= 2\pi i$$

$$\cdot \frac{1}{(-i-i) \left(-i - \left(\frac{\sqrt{3}}{3} + \frac{1}{2}\right)\right) \left(-i + \left(\frac{\sqrt{3}}{3} + \frac{1}{2}\right)\right) \left(-i - \left(\frac{\sqrt{3}}{3} - \frac{1}{2}\right)\right) \left(i - + \left(\frac{\sqrt{3}}{3} - \frac{1}{2}\right)\right)}$$

$$= 2\pi i \frac{1}{(-2i)(-1)(-3)}$$

$$= -\frac{\pi}{3}.$$

5

$$\int f(z)dz = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{x^2}{(x^2+1)(x^2+4)} dx + \int_{CR} f(z)dz.$$

$$\lim_{R \rightarrow \infty} \left| \int_{CR} f(z)dz \right| \leq \int_0^\pi \frac{|R^2 e^{2i\theta}|}{|(R^2 e^{2i\theta} + 1)| |(R^2 e^{2i\theta} + 4)|} dz \leq 0.$$

$$Res(f, i) + Res(f, 2i) = \lim_{R \rightarrow \infty} 2 \int_0^R \frac{x^2}{(x^2+1)(x^2+4)} dx$$

$$\begin{aligned} \int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx &= \frac{2\pi i \lim_{z \rightarrow i} (z-i) \frac{1}{(z-i)(z+i)(z-2i)(z+2i)}}{2} \\ &\quad + \frac{2\pi i \lim_{z \rightarrow 2i} (z-2i) \frac{1}{(z-i)(z+i)(z-2i)(z+2i)}}{2} \end{aligned}$$

$$\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx = \pi i \frac{1}{(2i)(i)(3i)} + \pi i \frac{1}{(i)(3i)(4i)}$$

$$\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx = \frac{\pi}{6} + \frac{\pi}{12}$$

$$\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx = \frac{\pi}{4}.$$

6

$$\begin{aligned} & \int_0^R \frac{1}{x^3+1} dx + \int_R^0 \frac{e^{\frac{2\pi i}{3}}}{(Re^{\frac{2\pi i}{3}})^3+1} dR \\ & + \int_0^{\frac{2\pi i}{3}} \frac{iRe^{i\theta}}{R^3e^{3i\theta}+1} d\theta = \int f(z) dz \end{aligned}$$

$$\begin{aligned} & \int_0^R \frac{1}{x^3+1} dx - e^{\frac{2\pi i}{3}} \int_0^R \frac{1}{R^3+1} dR + \int_0^{\frac{2\pi i}{3}} \frac{iRe^{i\theta}}{R^3e^{3i\theta}+1} d\theta \\ & = 2\pi i \lim_{z \rightarrow \frac{1}{2} - i\frac{\sqrt{3}}{2}} \left(z + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right) \frac{1}{\left(z + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right) \left(z + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \right) (z+1)} \end{aligned}$$

$$\begin{aligned} & \lim_{R \rightarrow \infty} \left(1 - e^{\frac{2\pi i}{3}} \right) \int_0^R \frac{1}{x^3+1} dx + \int_0^{\frac{2\pi i}{3}} \frac{iRe^{i\theta}}{R^3e^{3i\theta}+1} d\theta \\ & = \lim_{R \rightarrow \infty} 2\pi i \frac{1}{\left(\frac{1}{2} - i\frac{\sqrt{3}}{2} + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right) \left(\frac{1}{2} - i\frac{\sqrt{3}}{2} + 1 \right)} \end{aligned}$$

$$\begin{aligned} & \left(1 - \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right) \int_0^\infty \frac{1}{x^3+1} dx + \int_0^{\frac{2\pi i}{3}} \lim_{R \rightarrow \infty} \frac{iRe^{i\theta}}{R^3e^{3i\theta}+1} d\theta \\ & = 2\pi i \frac{1}{(-i\sqrt{3}) \left(\frac{3}{2} + i\frac{\sqrt{3}}{2} \right)} \end{aligned}$$

$$\int_0^{\infty} \frac{1}{x^3 + 1} dx = 2\pi i \frac{1}{\left(\frac{3}{2} - i\frac{\sqrt{3}}{2}\right) \left(\frac{3}{2} - i\frac{3\sqrt{3}}{2}\right)}$$

$$\int_0^{\infty} \frac{1}{x^3 + 1} dx = 2\pi i \frac{1}{\left(-i\frac{3\sqrt{3}}{4} + -i\frac{9\sqrt{3}}{4}\right)}$$

$$\int_0^{\infty} \frac{1}{x^3 + 1} dx = \frac{8\pi}{(3\sqrt{3} + 9\sqrt{3})}$$

$$\int_0^{\infty} \frac{1}{x^3 + 1} dx = \frac{2\pi}{3\sqrt{3}}$$

7

Since the equation is odd

$$\int_0^{\infty} \frac{x^2}{(1+x^2)x^2} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)x^2} dx.$$

Now consider region bounded by the union of a contour along the Real axis with distance to origin R and a positively oriented arc centered at 0 from the

positive real axis to the negative real axis with radius R . Then

$$\int_{-R}^R \frac{x^2}{(1+x^2)x^2} dx + \int_{CR} f(z) dz = \int_{Re \cup CR} f(z) dz$$

$$\lim_{R \rightarrow \infty} 2 \int_0^R \frac{x^2}{(1+x^2)x^2} dx = \lim_{R \rightarrow \infty} \sum_i \text{Res}(f, z_i) - \int_{CR} f(z) dz$$

$$\begin{aligned} 2 \int_0^\infty \frac{x^2}{(1+x^2)x^2} dx &= \frac{2\pi i}{1} \lim_{z \rightarrow z_0} \frac{d}{dz} \left((z-i)^2 \frac{z^2}{(z-i)^2(z+i)^2} \right) \\ &\quad - \lim_{R \rightarrow \infty} \int_0^\pi \frac{R^2 e^{2i\theta}}{(1+R^2 e^{2i\theta})^2} i R e^{i\theta} d\theta \end{aligned}$$

$$\begin{aligned} \int_0^\infty \frac{x^2}{(1+x^2)x^2} dx &= \pi i \lim_{z \rightarrow z_0} \left(\frac{2z}{(z+i)^2} - \frac{2z^2}{(z+i)^3} \right) \\ &\quad - \frac{i}{2} \int_0^\pi \lim_{R \rightarrow \infty} \frac{R^3 e^{3i\theta}}{(1+2R^2 e^{2i\theta} + R^4 e^{4i\theta})} d\theta \\ &= \pi i \left(\frac{-2i}{4} + \frac{2i}{8} \right) - \frac{i}{2} \int_0^\pi 0 d\theta \\ &= \frac{\pi}{4}. \end{aligned}$$